

ASSIGNMENT 1

PROBABLITY

1. Identify which of these form a σ -algebra:
 - (a) $\Omega = \{1, 2, 3\}$ and $\mathcal{F} = \{\emptyset, \Omega\}$.
 - (b) $\Omega = \{1, 2, 3\}$ and $\mathcal{F} = \{\emptyset, \{1\}, \Omega\}$.
 - (c) $\Omega = \{1, 2, 3, 4\}$ and $\mathcal{F} = \{\emptyset, \Omega, \{1, 2\}, \{3, 4\}\}$.
 - (d) $\Omega = \{1, 2, 3, 4\}$ and $\mathcal{F} = \{\emptyset, \Omega, \{1, 2\}, \{2, 3\}, \{1, 2, 3\}\}$.
2. Let $(\Omega, \mathcal{F})$ be a measurable space. Show that:
 - (a) $\Omega \in \mathcal{F}$
 - (b) If $(A_n)_{n=1}^k$ is a finite sequence of $\mathcal{F}$ -measurable sets, then $\bigcup_{n=1}^k A_n \in \mathcal{F}$.
 - (c) If $(A_n)_{n=1}^\infty$ is a sequence of $\mathcal{F}$ -measurable sets, then $\bigcap_{n=1}^\infty A_n \in \mathcal{F}$.
3. Let $\mathcal{A}, \mathcal{A}_1, \mathcal{A}_2 \subseteq 2^\Omega$ be collections of subsets of Ω . Show that:
 - (a) If $\mathcal{A}_1 \subset \mathcal{A}_2$, then $\mathcal{F}(\mathcal{A}_1) \subset \mathcal{F}(\mathcal{A}_2)$.
 - (b) If $\mathcal{A}$ is a σ -field, then $\mathcal{F}(\mathcal{A}) = \mathcal{A}$.
 - (c) $\mathcal{F}(\mathcal{F}(\mathcal{A})) = \mathcal{F}(\mathcal{A})$.
 - (d) If $\mathcal{A}_1 \subset \mathcal{F}(\mathcal{A}_2)$, then $\mathcal{F}(\mathcal{A}_1) \subset \mathcal{F}(\mathcal{A}_2)$.
4. Let $\Omega = \{1, 2, 3, 4, 5, 6\}$ and $\mathcal{A} = \{\{1, 3, 5\}\}$. Find $\sigma(\mathcal{A})$.
5. What is the interpretation of the 2^Ω σ -algebra?
6. Can there be a σ -field with 6 elements in it? If yes, give an example, else state why not.
7. Let $\Omega = \{1, 2, 3, 4\}$. Find four distinct σ -fields $(\mathcal{F}_i)_{i=1}^4$ such that $\mathcal{F}_i$ forms a filtration, i.e., $\mathcal{F}_1 \subset \mathcal{F}_2 \subset \mathcal{F}_3 \subset \mathcal{F}_4$.
8. Construct a $\Omega, \mathcal{F}_1, \mathcal{F}_2, \mathcal{F}_3$ where Ω is the sample space and three σ -fields are defined on it, such that $\mathcal{F}_1 \cup \mathcal{F}_2 \cup \mathcal{F}_3$ is not a σ -field.

9. Let $\Omega = \{1, 2, 3, 4, 5, 6\}$. Let $\mathcal{F}$ denote the σ -field generated by $\{2, 3, 4\}$ and $\{4, 5, 6\}$. Given $\mathbb{P}(\{1\}) = 1/6$, $\mathbb{P}(\{2, 3\}) = 1/4$, and $\mathbb{P}(\{4\}) = \mathbb{P}(\{5, 6\})$, find $\mathbb{P}(\{4\})$.
Let $X = 1_{\{2\}} - 1_{\{3,4\}} + 3 \times 1_{\{5,6\}}$. Is X $\mathcal{F}$ -measurable? Is X^2 $\mathcal{F}$ -measurable? Find $\mathbb{P}[X^4 \leq 2]$.
10. Let $\Omega = \{1, 2, 3, 4, 5, 6\}$, $\mathcal{F} = 2^\Omega$ with $\mathbb{P}(\{1\}) = \mathbb{P}(\{2\}) = 1/12$, and $\mathbb{P}(\{3\}) = \mathbb{P}(\{4\}) = 1/4$. Find $\mathbb{P}(\{5\})$ and $\mathbb{P}(\{6\})$ if the events $\{1, 3, 4\}$ and $\{1, 2, 3, 5\}$ are independent.
11. Let $\Omega = \{1, 2, 3, 4, 5, 6\}$, $\mathcal{F} = 2^\Omega$ and $\mathbb{P}(\{1\}) = \mathbb{P}(\{2\}) = \mathbb{P}(\{3\}) > 0$, $\mathbb{P}(\{4\}) = \mathbb{P}(\{5\})$, and $\mathbb{P}(\{6\}) = 1/12$. If $X = 1_{\{1,2\}} - 1_{\{3,4\}}$ and $Y = 2 \times 1_{\{1,3,5\}} + 3 \times 1_{\{2,4,6\}}$ are independent random variables, find $\mathbb{P}(\{1\})$ and $\mathbb{P}(\{4\})$.
12. Construct a random variable X on an Ω with two elements such that $\mathbb{E}[X^2] \neq (\mathbb{E}[X])^2$.
13. Let $\Omega = \{1, 2, 3, 4, 5, 6\}$, $\mathcal{F} = 2^\Omega$; $\mathbb{P}(\{i\}) = 1/12$ for $i \leq 3$ and $\mathbb{P}(\{i\}) = a/i$ for $i > 3$. Let $X(i) = i^2$ for $i \leq 4$ and $X(5) = X(6) = 2$. If $A = \{1, 2, 5\}$ and $g(x) = x^3$ for $x \in \mathbb{R}$, find $\int_A g(x) d\mathbb{P}$ and $\mathbb{E}[g(X)]$.
14. Let $\Omega = \{1, 2, \dots, 10\}$, $\mathcal{F} = 2^\Omega$. If $\mathbb{P}(\{i\}) = i^2/a$ for all i and $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space, find a .
Let $X(i) = 1$ for $i \leq 5$ and $X(i) = i$ for $i \geq 6$. Let $Y = 2 \times 1_{\{1,2,3\}} + 3 \times 1_{\{4,5,6\}}$. Find $\mathbb{E}[X]$, $\mathbb{E}[XY]$, $\mathbb{E}[Y|X]$, $\mathbb{E}[X|Y^2]$, $\mathbb{E}[XY|\sigma(X)]$, and $\mathbb{E}[\mathbb{E}[X|\sigma(Y)]|\sigma(X)]$.